

Encoding fMRI Responses with Large Language Models in a Naturalistic Listening Task (Team-9, A1)

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Problem Statement

Existing Work

Dataset Overview: **Le Petit Prince Multi-Talker Dataset (2025)**

- Naturalistic multimodal neuroimaging dataset
- Contains 7T fMRI and EEG recordings
- Based on the Chinese audiobook version of Le Petit Prince
- Chinese translation of Le Petit Prince
- 26 native Mandarin Chinese speakers [26 Subjects]
- Narrated using synthetic male and female voices

Dataset Acquisition Setup:

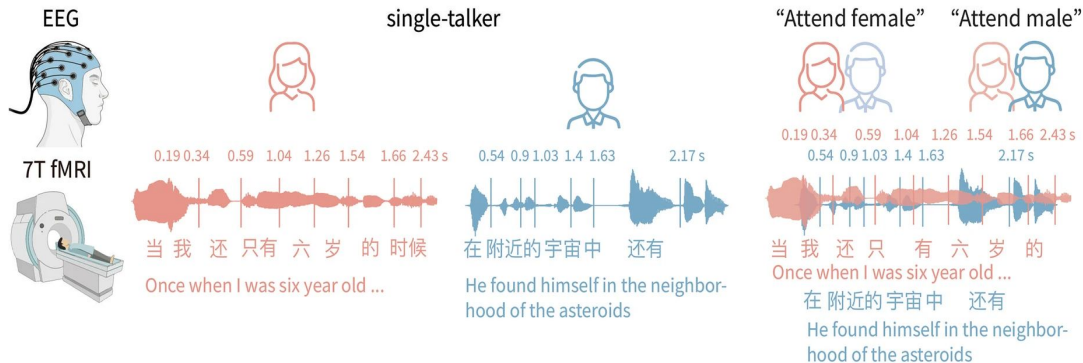
Listening Conditions

1. Single-speaker - female only (Run 1)
2. Single-speaker - male only (Run 2)
3. Multi-speaker: attend female (Run 3)
4. Multi-speaker: attend male (Run 4)

Multi-Talker Setup

- Male and female speech streams played simultaneously
- Participants instructed which speaker to attend to

a Experiment procedures



b Acoustic annotations

time	pitch	intensity
...	...	...
0.02	179.31	64.73
0.03	349.96	70.85
0.04	341.81	73.64
...	...	...

c Word annotations

word	onset	offset	duration	logfreq	pos	*td	*bu	*lc
当 在	0 0.25	0.19 0.54	0.19 0.29	8.53 11.76	ADP ADP	3 3	1 1	2 2
我 附近	0.19 0.54	0.34 0.90	0.15 0.36	10.53 6.46	PRON NOUN	5 5	2 2	3 3
还 的	0.34 0.90	0.59 1.03	0.29 0.13	9.87 13.55	ADV PART	3 1	2 2	3 2
只有 宇宙	0.67 1.03	1.04 1.40	0.37 0.37	8.51 5.87	ADV NOUN	2 2	1 3	2 3
...	...	...	...	...	...	...	...	...

*Node counts from three parsing strategies: td: top-down, bu: bottom-up, lc: left-corner

Data Preprocessing

They convert Dicom MRI images to NIfTI format

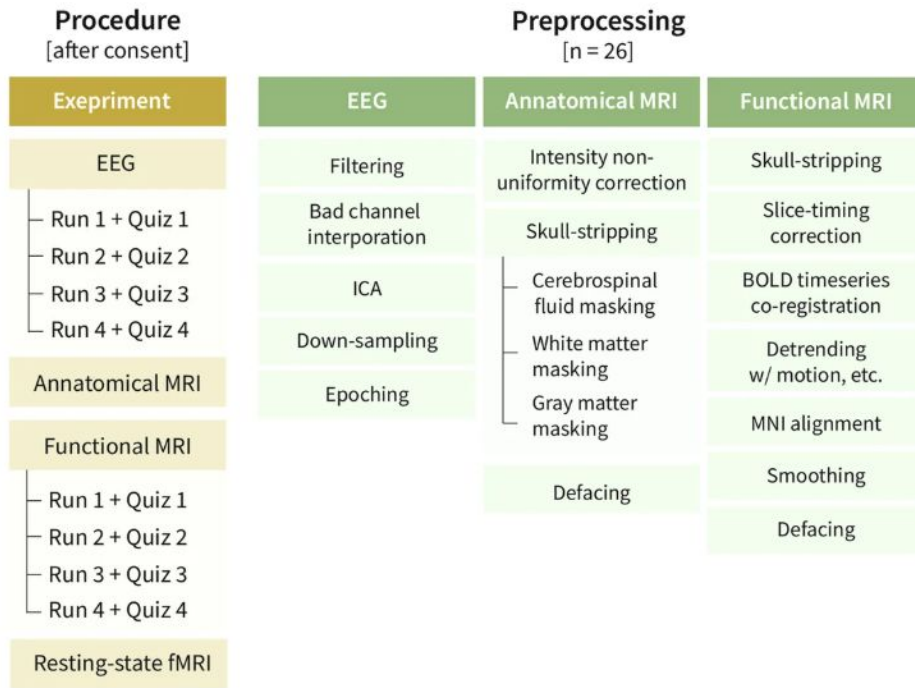
Motion Outlier Detection

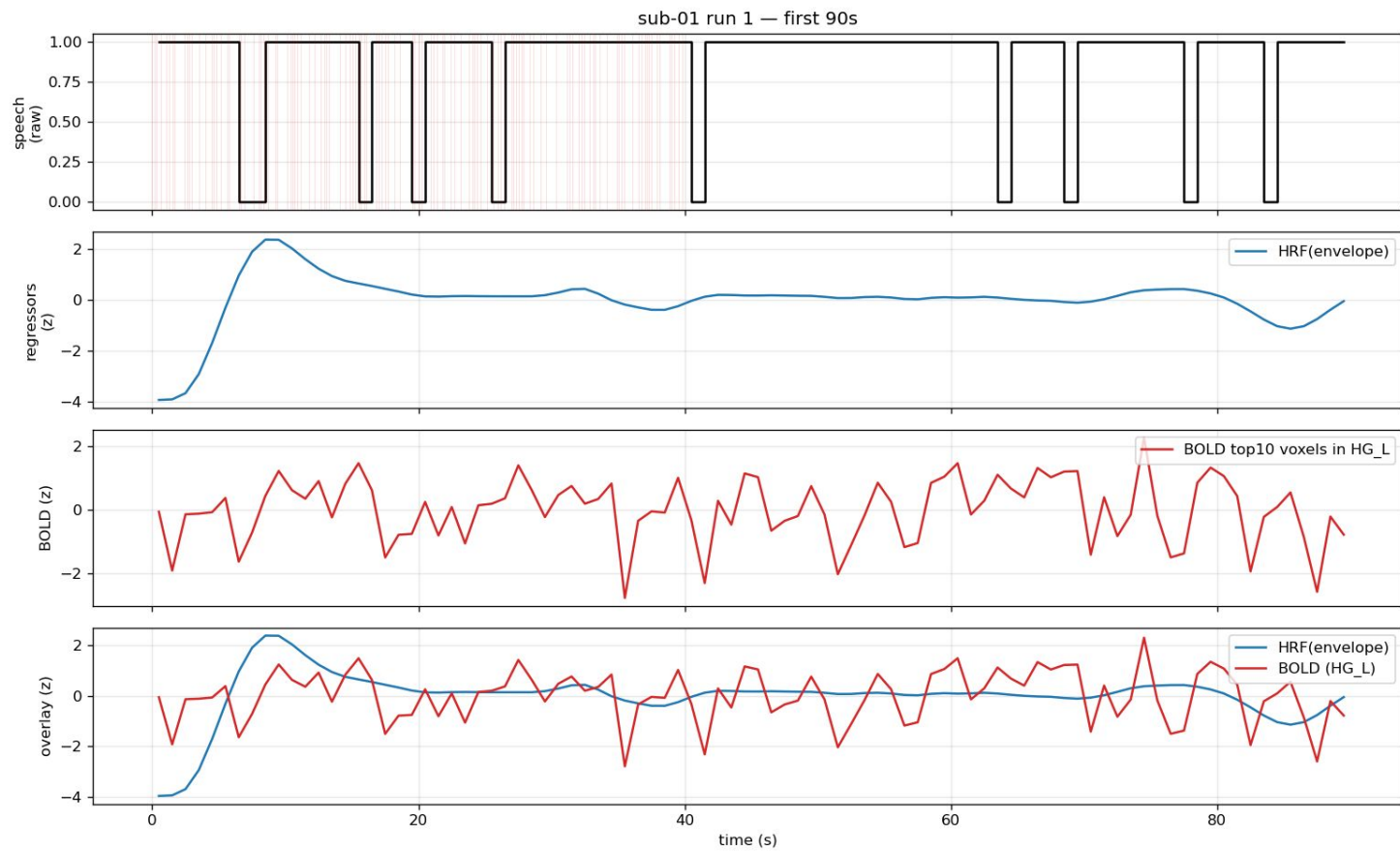
Volumes flagged when:

- $FD > 0.5 \text{ mm}$
- Standardized DVARS > 1.5

Final Output

- Preprocessed BOLD signals in MNI space
- To be used for Inter Subject Correlation (ISC), encoding, and GLM analyses





Evaluation Metrics

1. Pearson Correlation (r):

Is the similarity between predicted fMRI responses from the model, and actual recorded brain activity.

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$

2. 2v2 Accuracy :

tests if the model can distinguish between two repetitions of one stimulus (a1, a2) and two repetitions of another (b1, b2).

Given:

- Predicted representations: $\hat{y}_1, \hat{y}_2$
- Ground-truth brain responses: y_1, y_2

The prediction is considered correct if:

$$d(\hat{y}_1, y_1) + d(\hat{y}_2, y_2) < d(\hat{y}_1, y_2) + d(\hat{y}_2, y_1)$$

Experimentation Details

Models Evaluated

Qwen3 1B,4B; Llama3 1.3B,3B; Llada 8B-Instruct

Defined Experiments

EXPERIMENT 01

Top 7 Language ROIs

Analysis of primary language processing regions in the brain.

EXPERIMENT 02

31 Harvard-Oxford ROIs (HO31)

Comprehensive evaluation across the standardized HO31 atlas.

EXPERIMENT 03

Top 10% ISC Voxel Selection

Focus on voxels showing highest inter-subject correlation.

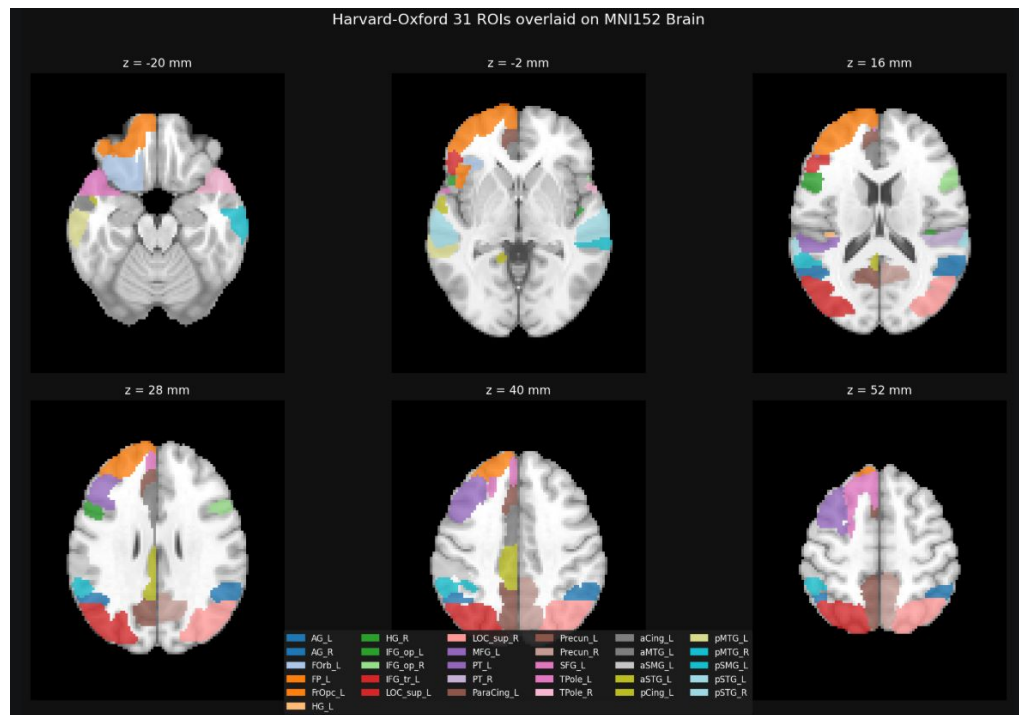
1. Top 7 Language ROIs

- Models compared: Qwen3-1.7B, Llama-3.2-1B, LLaDA-8B-Instruct
- Protocol: (Cross-subject) train on 21 subjects, test on held-out 3
- Ridge regression ($\alpha = 300$) maps LLM hidden states to BOLD per voxel

ROI	Region
TP	Temporal Pole
aSTS	Anterior Superior Temporal Sulcus
pSTS	Posterior STS
BA44	Broca's area (pars opercularis)
BA45	Broca's area (pars triangulis)
BA47	Inferior Frontal Gyrus
AG_TPJ	Angular Gyrus

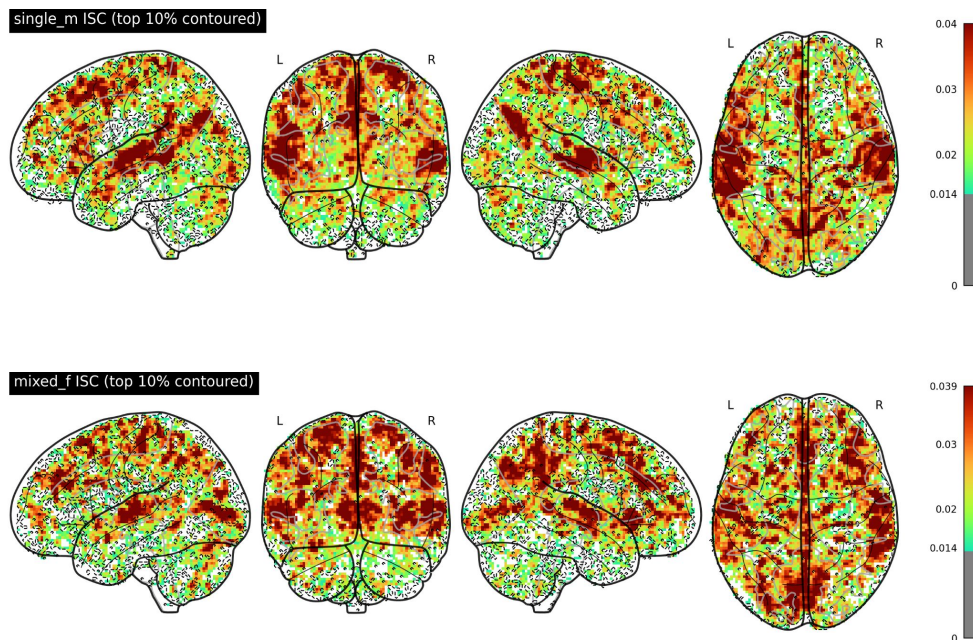
2. 31 Harvard-Oxford ROIs (HO31)

- 31 bilateral ROIs spanning frontal, temporal, parietal, and occipital cortex
- Models: Qwen3-4B and Llama-3.2-3B (base + fine-tuned)
- Alpha sweep ($\alpha \in \{10, 300, 1000\}$) run to find optimal ridge regularization across the larger feature space
- Noise ceiling computed as LOO-ISC per ROI



3. Top 10% ISC Voxel Selection

- Inter-Subject Correlation
- Leave One Out ISC per voxel using training subjects only
- Keep only the top 10% voxels as the prediction target.



Noise Ceiling Analysis

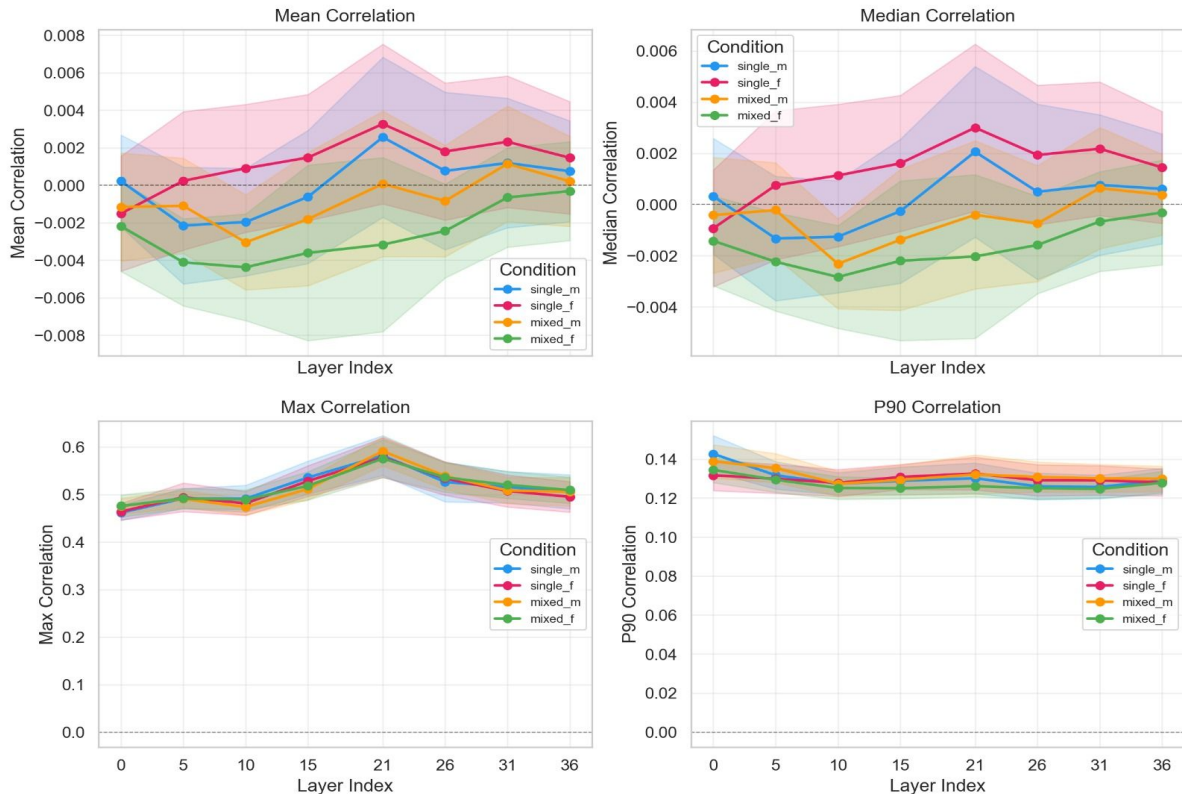
The noise ceiling (NC) is the theoretical upper bound on how well any encoding model could predict BOLD responses.

If the model's encoding correlation approaches NC $\rightarrow$ the model is extracting nearly all the predictable neural variance.

$$NC = \frac{1}{N} \sum_{i=1}^N \text{mean_corr}(\text{BOLD}_i, \overline{\text{BOLD}}_{-i})$$

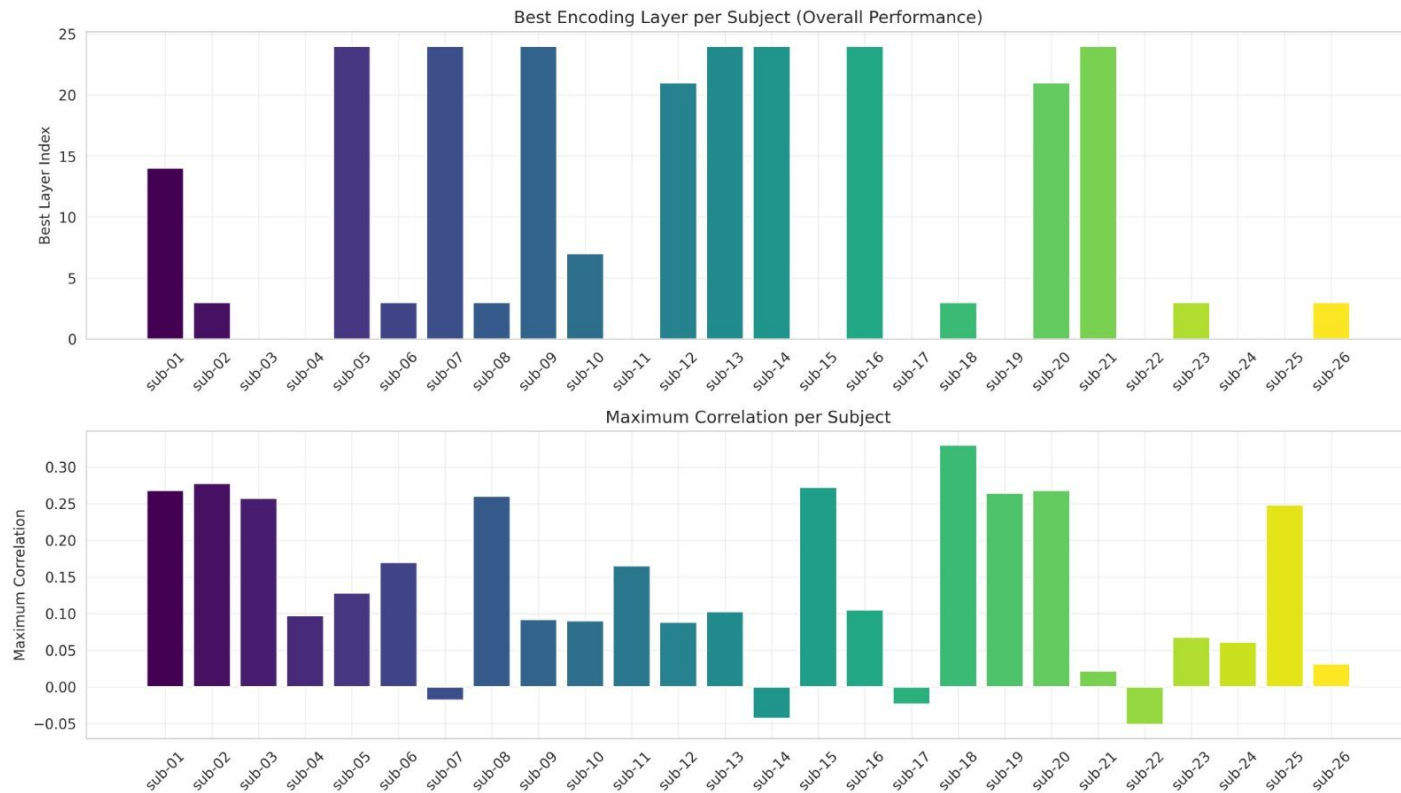
Baseline: Qwen3 Inference on Entire Dataset voxels (8,53,724 voxels)

Qwen3-8B — Layer-wise Encoding Performance by Condition (split=train)

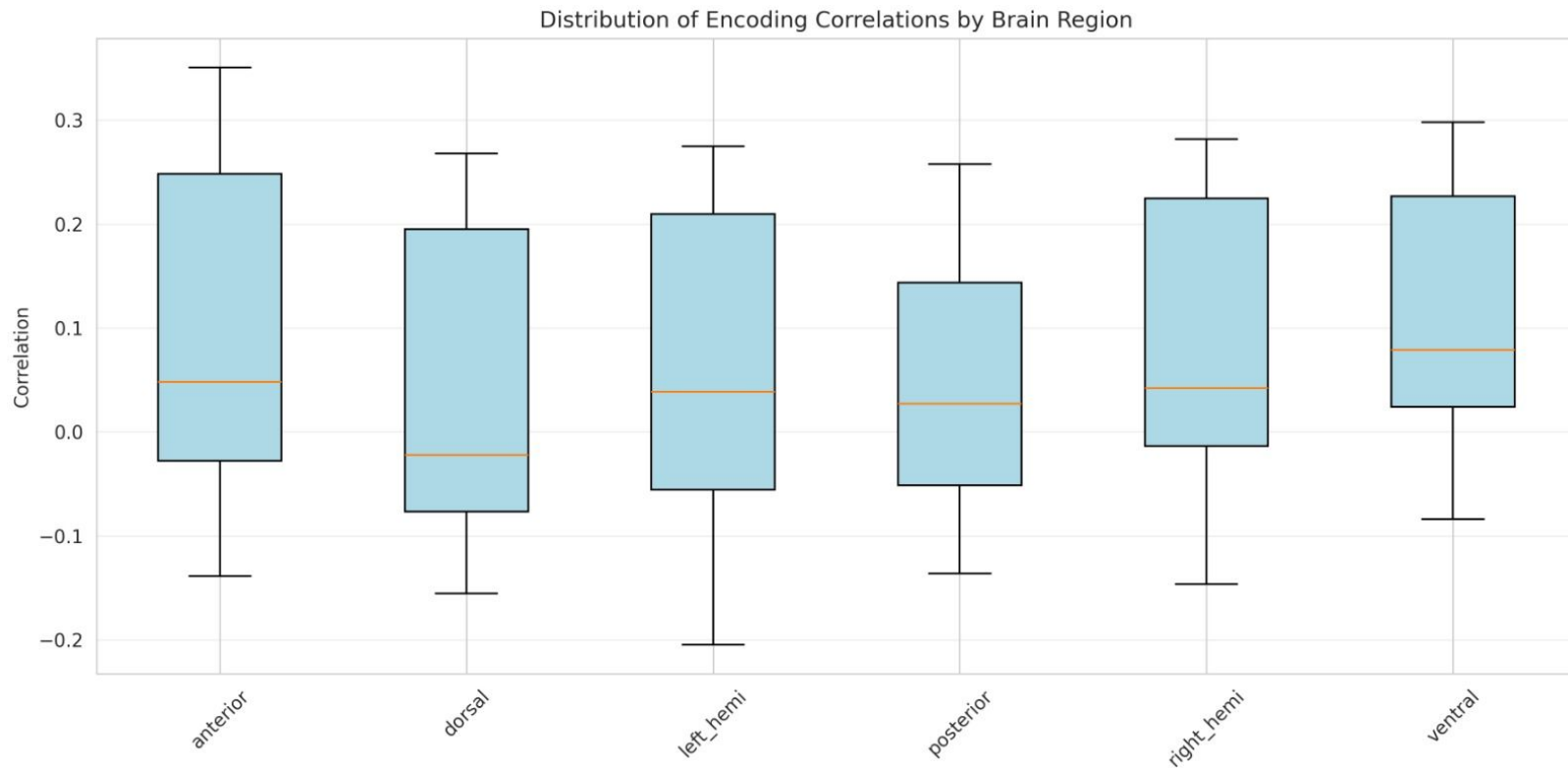


Results

Qwen: Performance on Every Subject

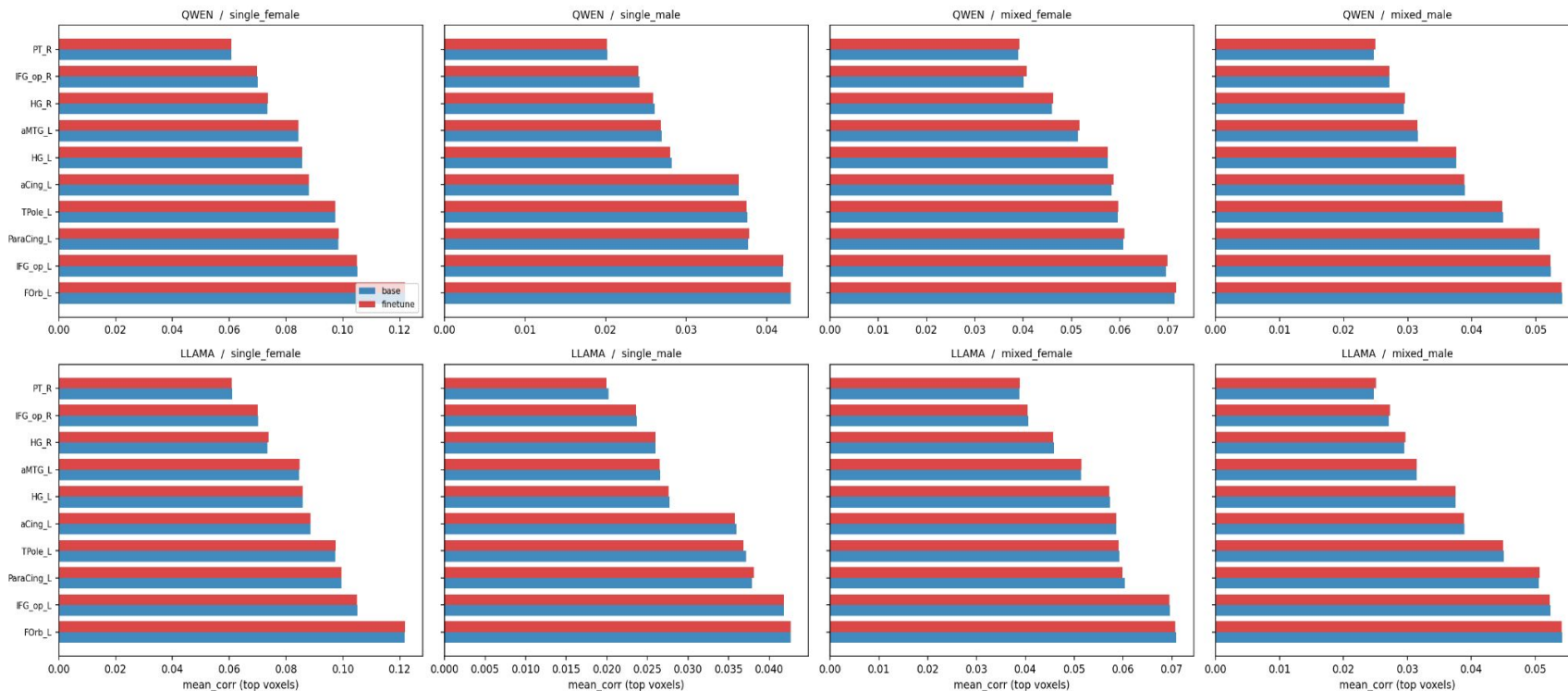


Qwen: Mean Correlation on different lobes

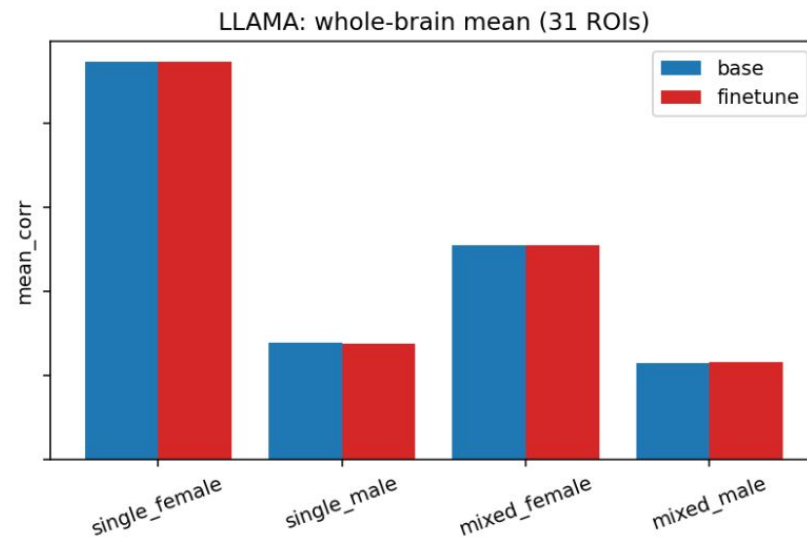
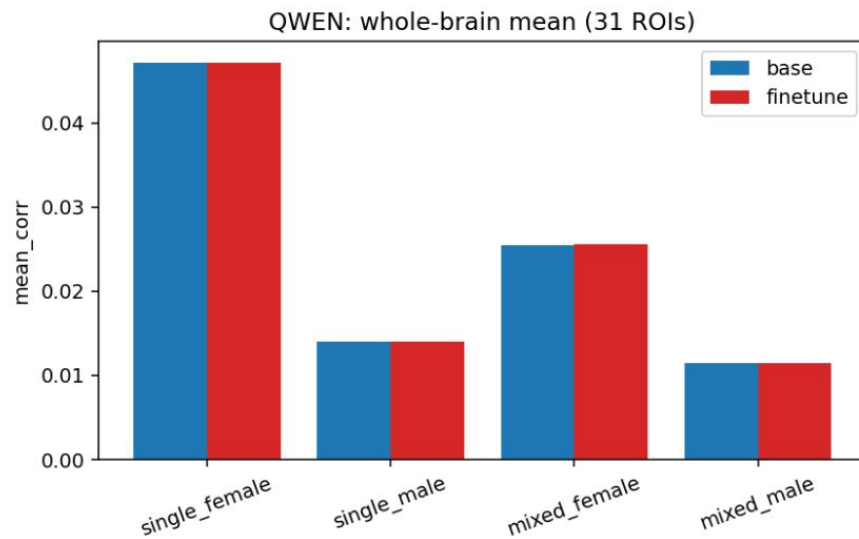


Top 10 ROIs - Base vs Finetune

Base vs Fine-tune — top-10 ROIs per condition

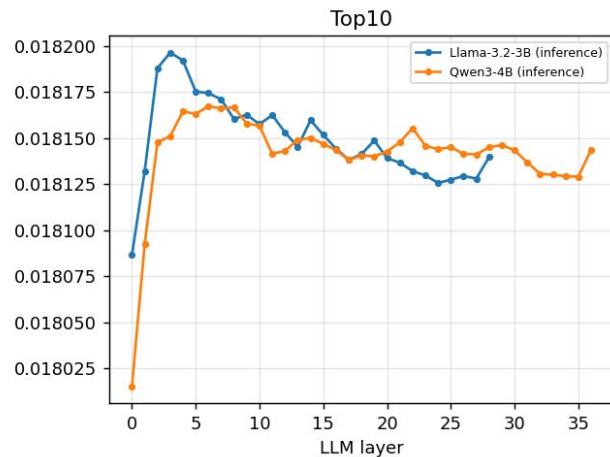
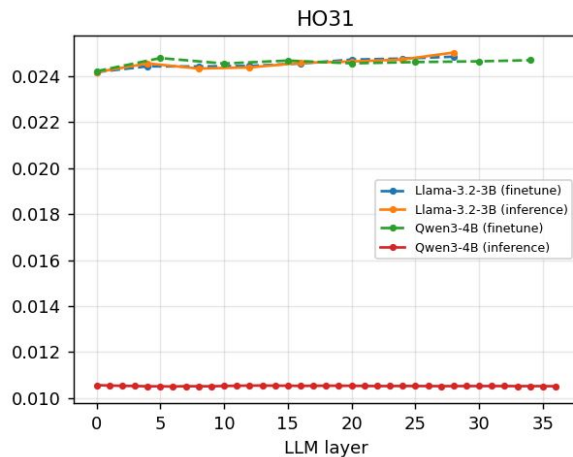
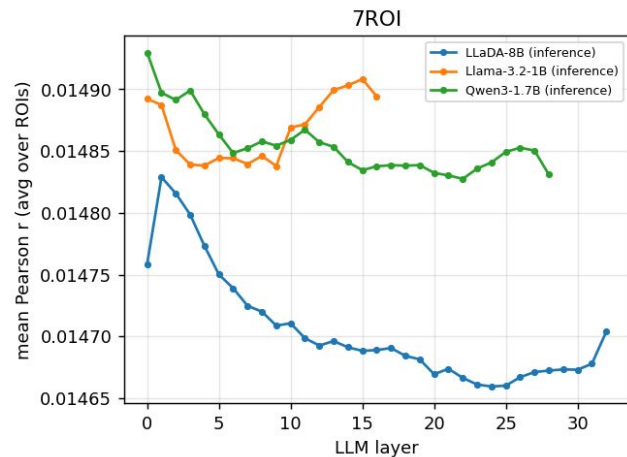


31 ROIs - Base vs Finetune mean correlation



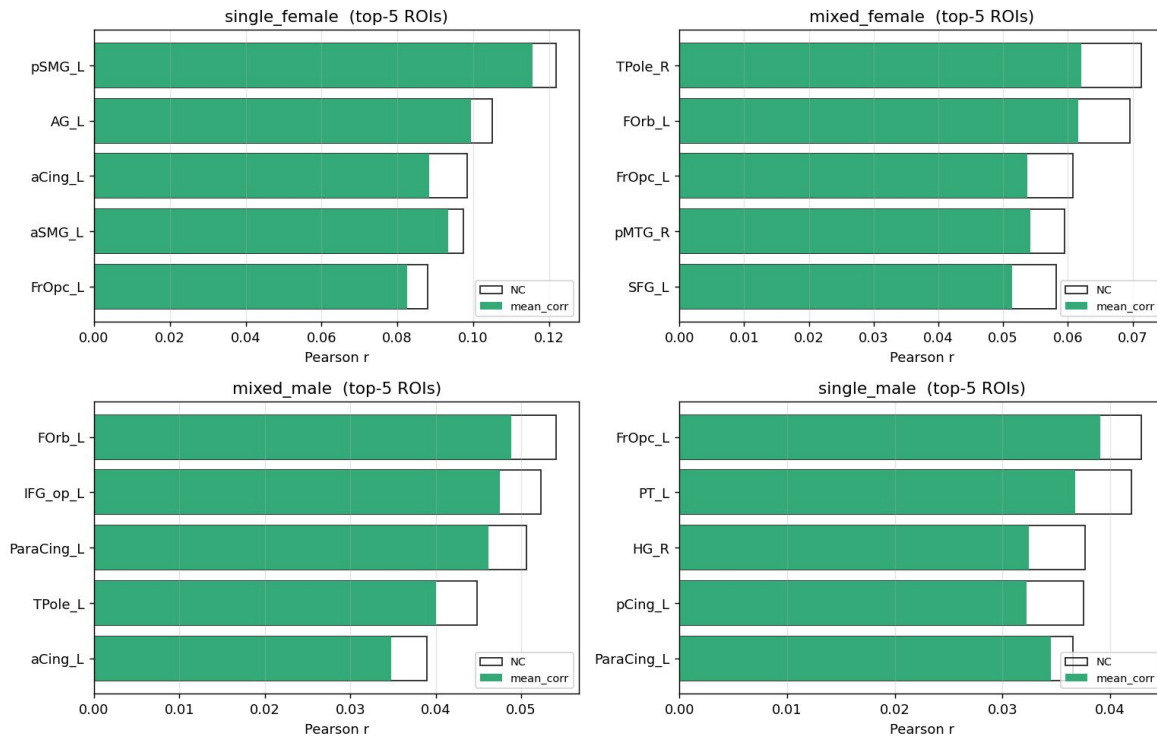
Encoding correlation per LLM layer

Encoding correlation per LLM layer

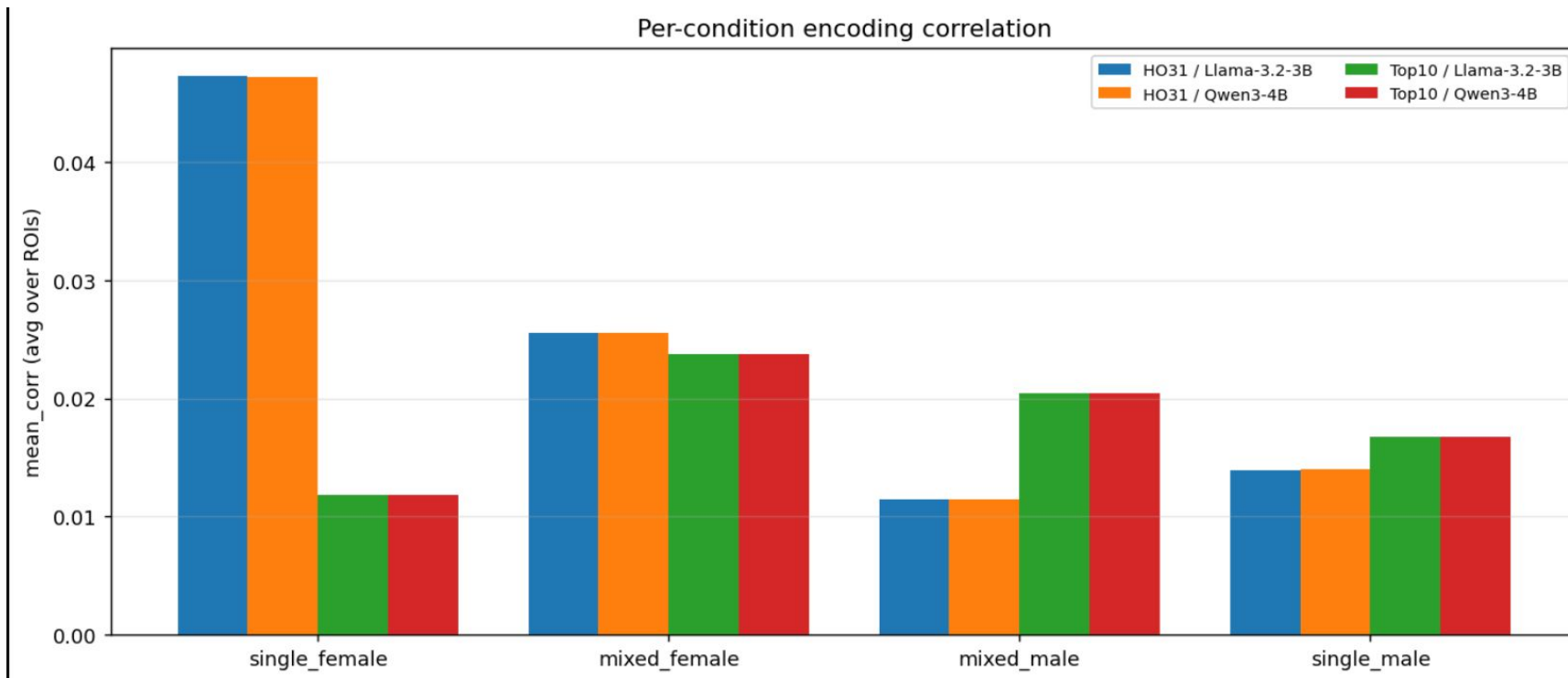


31 ROIs - Top 5 ROIs Mean Correlation vs Noise Ceiling

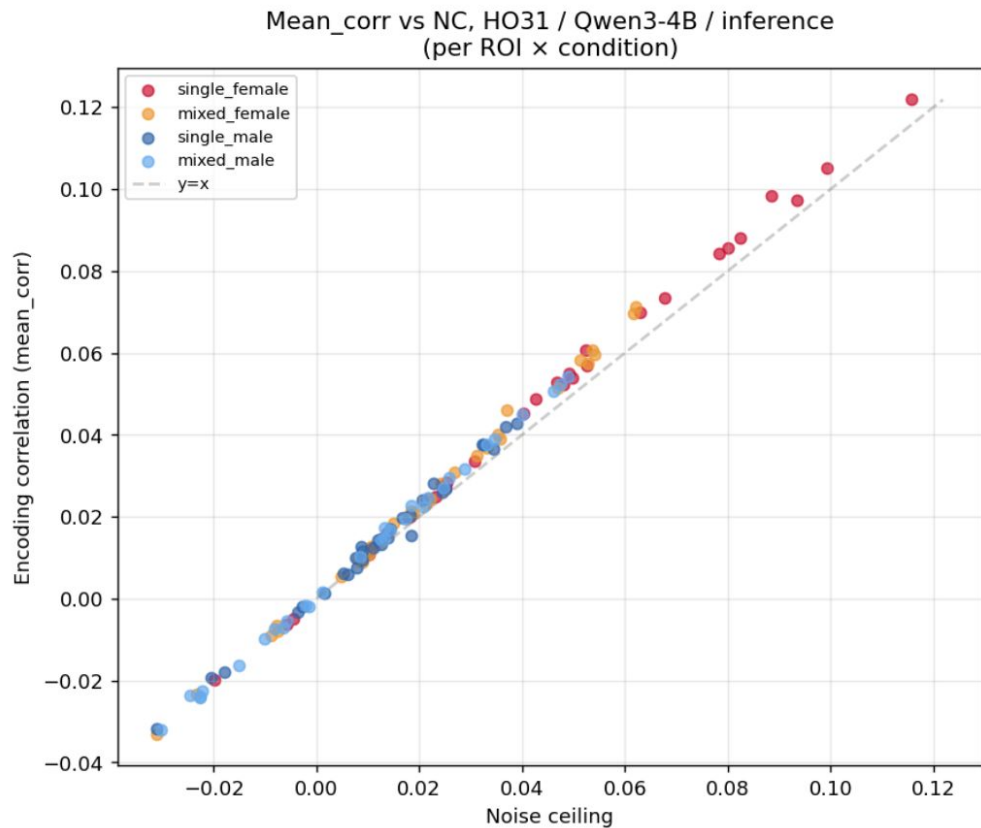
Top ROIs per condition (HO31 / Qwen3-4B / inference)



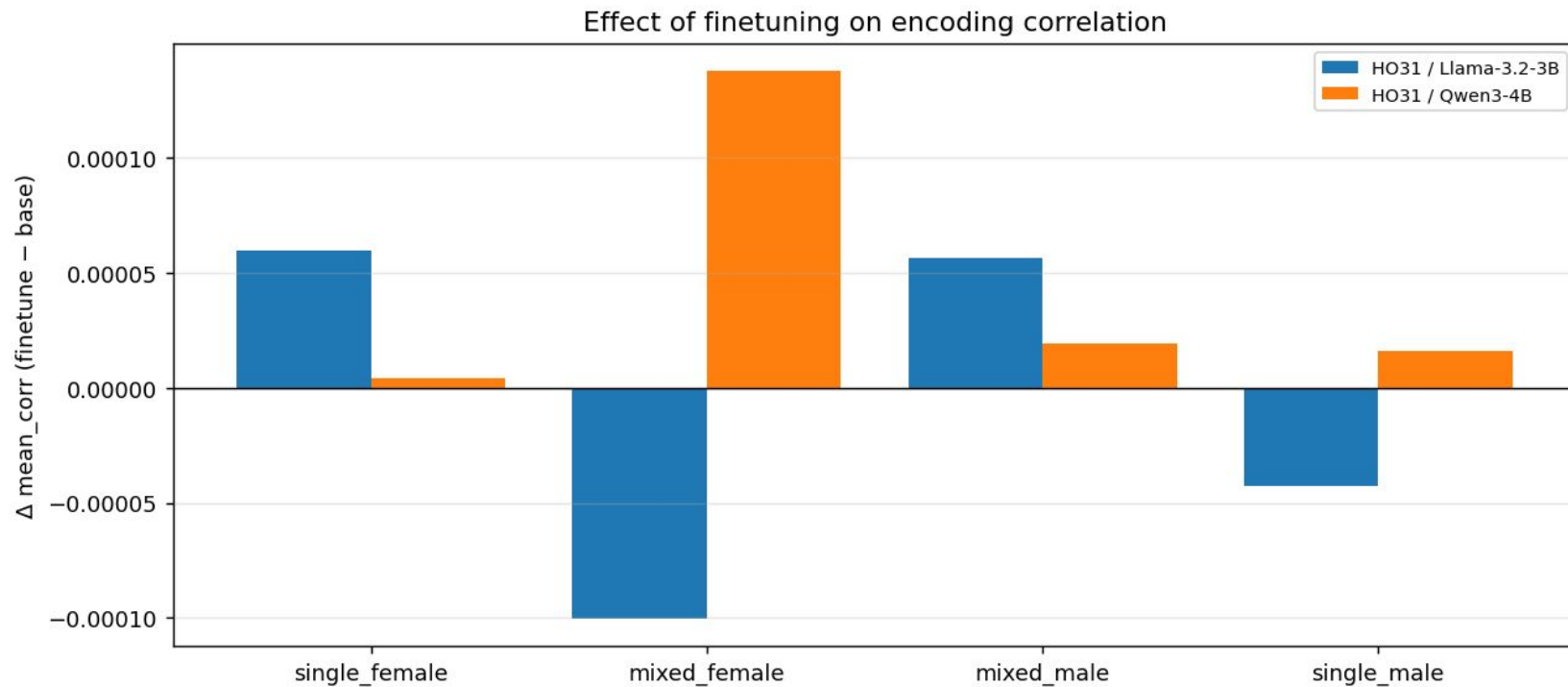
Per Experiment Encoding correlation Comparision



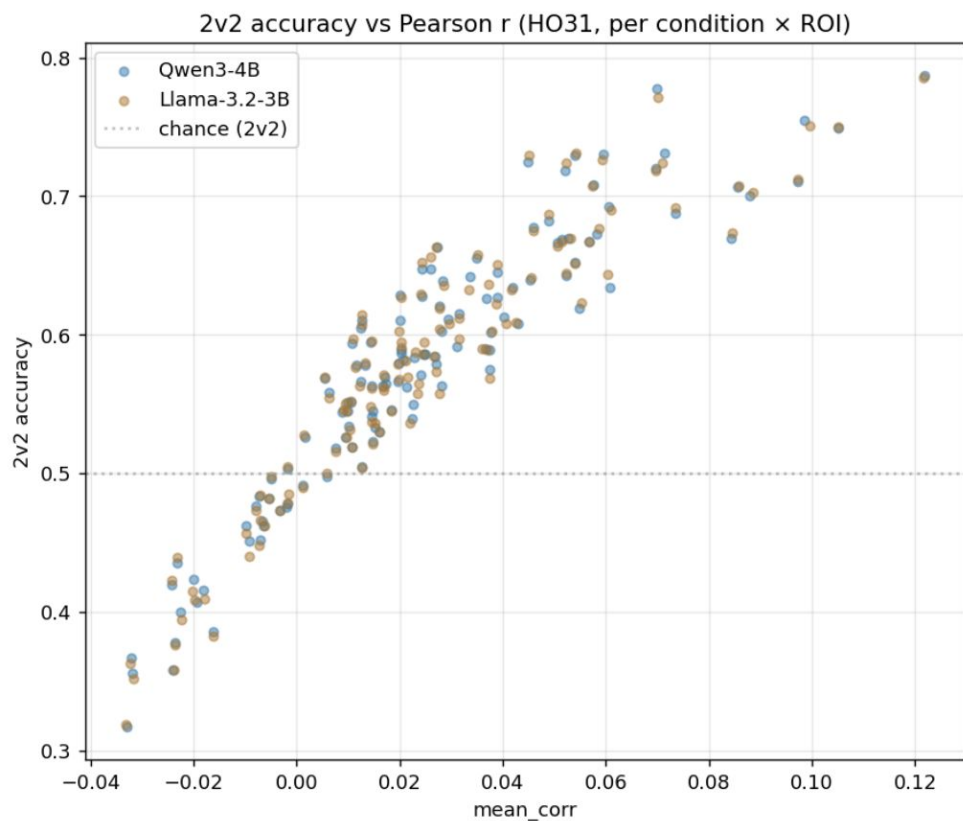
Qwen : Mean Correlation vs Noise Ceiling



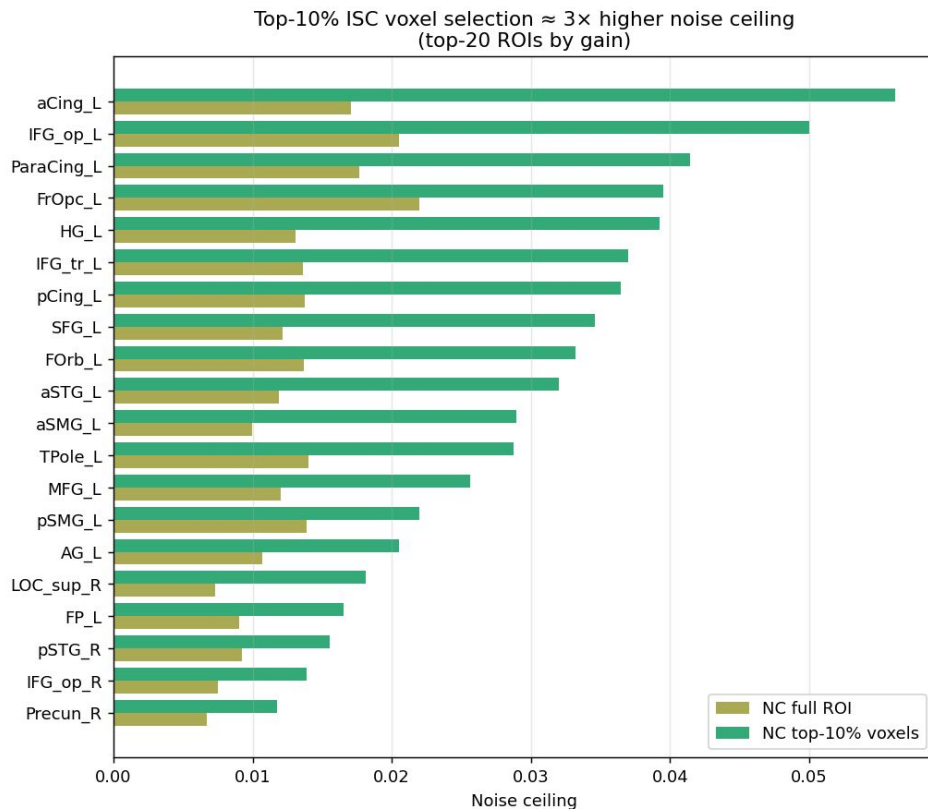
Finetuning Correlation

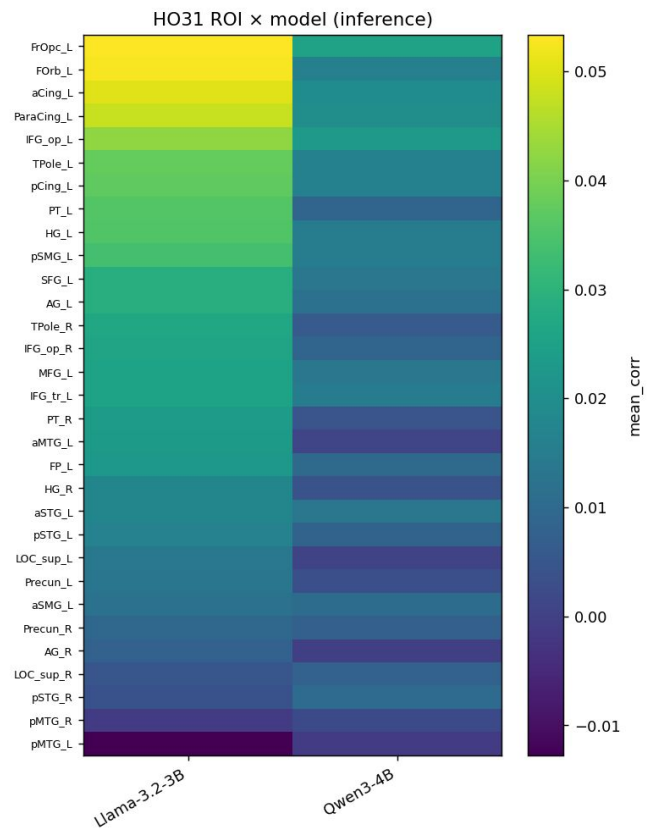


HO31: Pearson correlation vs 2v2 Accuracy

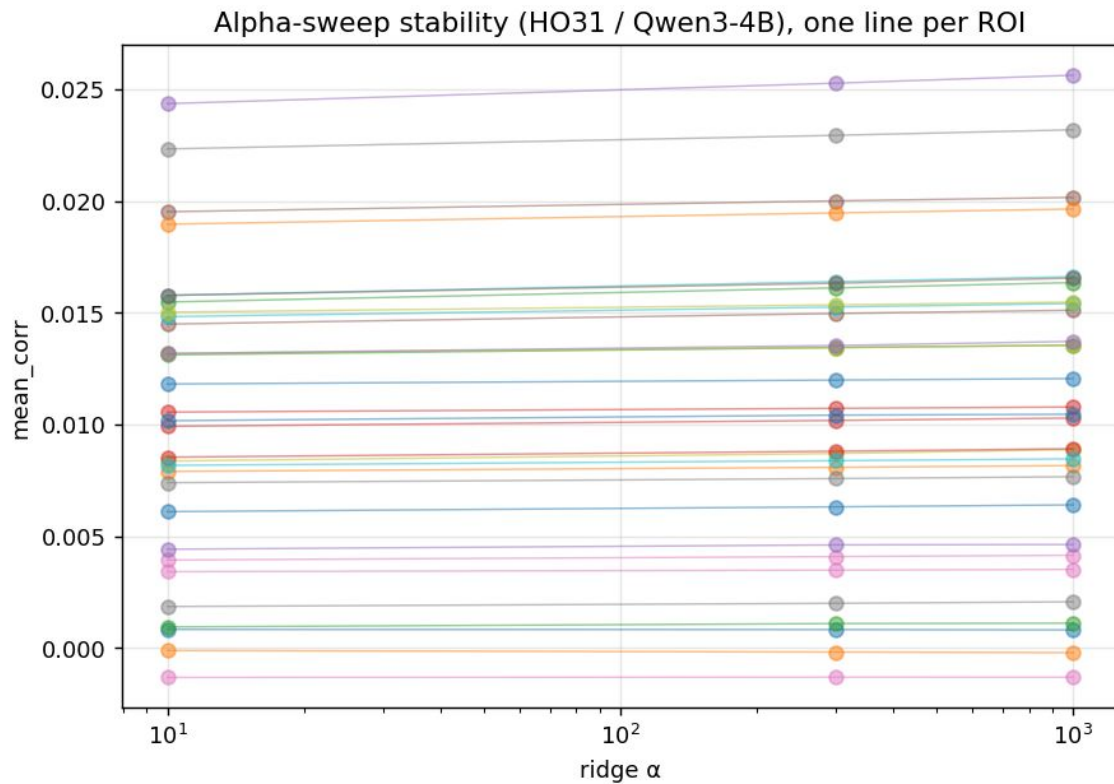


Noise Ceiling: All ROIs vs Top 10% ROIs





Variable alpha values Ablation [ridge regression regularization strength]



Results Overview

Table 1: Headline results (cross-subject evaluation).

Experiment	Model	Mode	n ROIs	mean_corr	2v2	NC	JS
7ROI	Qwen3-1.7B	inference	7	0.0149	0.510	—	—
7ROI	Llama-3.2-1B	inference	7	0.0149	0.510	—	—
7ROI	LLaDA-8B	inference	7	0.0148	0.510	—	—
HO31	Qwen3-4B	inference	31	0.0245	0.577	0.0297	0.43
HO31	Qwen3-4B	finetune	31	0.0246	0.577	0.0297	0.43
HO31	Llama-3.2-3B	inference	31	0.0245	0.577	0.0297	0.43
HO31	Llama-3.2-3B	finetune	31	0.0245	0.577	0.0297	0.43
Top10	Qwen3-4B	inference	4	0.0182	0.804	0.0186	—
Top10	Llama-3.2-3B	inference	4	0.0182	0.805	—	—
Top10	Llama-3.2-3B	finetune	4	0.0182	0.805	—	—

Mean Performance across top-5 ROIs of all experiments

Table 2: Mean performance across top-5 ROIs.

Experiment	Model	mean_corr (top5)	2v2 (top5)	NC (top5)	JS (top5)
7ROI	Qwen3-1.7B	0.040	0.574	—	—
7ROI	Llama-3.2-1B	0.038	0.569	—	—
7ROI	LLaDA-8B	0.051	0.589	—	—
HO31	Qwen3-4B (inf)	0.049	0.644	0.054	0.43
HO31	Qwen3-4B (FT)	0.049	0.644	0.054	0.43
HO31	Llama-3.2-3B (inf)	0.049	0.645	0.054	0.43
HO31	Llama-3.2-3B (FT)	0.049	0.645	0.054	0.43
Top10	Qwen3-4B	0.018	0.804	0.019	—
Top10	Llama-3.2-3B	0.018	0.805	0.019	—

Best Layers per Model and Experiment

Table 3: Layer-wise highlights (best layer and range covered).

Experiment / Model	Best Layer Information
7-ROI Qwen3-1.7B	Best layer 0 (29 layers, 0–28)
7-ROI Llama-3.2-1B	Best layer 15 (17 layers)
7-ROI LLaDA-8B	Best layer 1 (33 layers, 0–32)
HO31 Qwen3-4B	Best layer 0 of 37
HO31 Llama-3.2-3B	Best layer 28 of 29
Top10 Qwen3-4B	Best layer 6
Top10 Llama-3.2-3B	Best layer 3

Table 3: HO31 top-5 ROI results across listening conditions.

Condition	mean_corr	2v2	NC
All (avg) – previous table	0.049	0.644	0.054
Single Female	0.102	0.740	0.106
Single Male	0.039	0.605	0.045
Mixed Female	0.064	0.698	0.067
Mixed Male	0.048	0.682	0.052

Best ROIs per Experiment

Table 4: Best ROIs per experiment (top-5).

Experiment	Model	Top ROIs
7ROI	Qwen3-1.7B	BA45, aSTS, AG_TPJ, pSTS, BA47
7ROI	Llama-3.2-1B	BA45, aSTS, BA47, AG_TPJ, pSTS
7ROI	LLaDA-8B	BA45, aSTS, AG_TPJ, BA47, TP
HO31	Qwen3-4B (inf, boost)	FrOpc_L, FOrb_L, aCing_L, ParaCing_L, IFG_op_L
HO31	Qwen3-4B (FT)	FrOpc_L, FOrb_L, aCing_L, ParaCing_L, IFG_op_L
HO31	Llama-3.2-3B (both)	FrOpc_L, FOrb_L, aCing_L, ParaCing_L, IFG_op_L
Top10	Qwen3-4B	mixed_f_top10 > mixed_m > single_m > single_f
Top10	Llama-3.2-3B	same ordering

Results Summary